1. Questions

Study the following information carefully and answer the questions given below.

The given table shows the total number of books (fiction and non-fiction) sold by four different stores namely A, B, C, and D along with the number of fiction and non-fiction books sold by each store. Some data is missing in the table, which you need to calculate based on the given notes.

Store	The total number of books sold	The number of fiction books sold	The number of non-fiction books sold
A	500	-	-
B	-	250	-
C	450	-	250
D	-	-	300

Notes:

1). The ratio of Fiction to Non-Fiction books sold by store B is 5:7.

2). The total number of books sold by store D is 25% more than the total number of books sold by store B.

3). The number of Non-Fiction books sold by store A is 20% less than the number of Non-Fiction books sold by store C.

Find the ratio of the number of Fiction books sold by store A to the number of Non-Fiction books sold by store D.

a. 4 : 5

b. 3 : 1

c. 2 : 3

d. 1 : 1

e. 3 : 4

2. Questions

The total number of books sold by store C is what percentage of the number of non-fiction books sold by store A?

a. 210%

b. 225%

c. 220%

d. 280%

e. 245%

3. Questions

The average number of books sold by stores A and B together is how much more or less than the average number of non-fiction books sold by stores B and C together?

a. 230 more

b. 250 less

c. 200 more

d. 180 less

e. 250 more

4. Questions

If the number of fiction books sold by store B is x% more than that by store C, find the value of x.

a. 32

b. 20

c. 25

d. 38

e. 35

5. Questions

If the total number of books sold by store E is 20% more than that by store D and the number of non-fiction books sold by store E is 50 more than that by store B. Find the number of fiction books sold by store E.

a. 500

b. 300

c. 400

d. 200

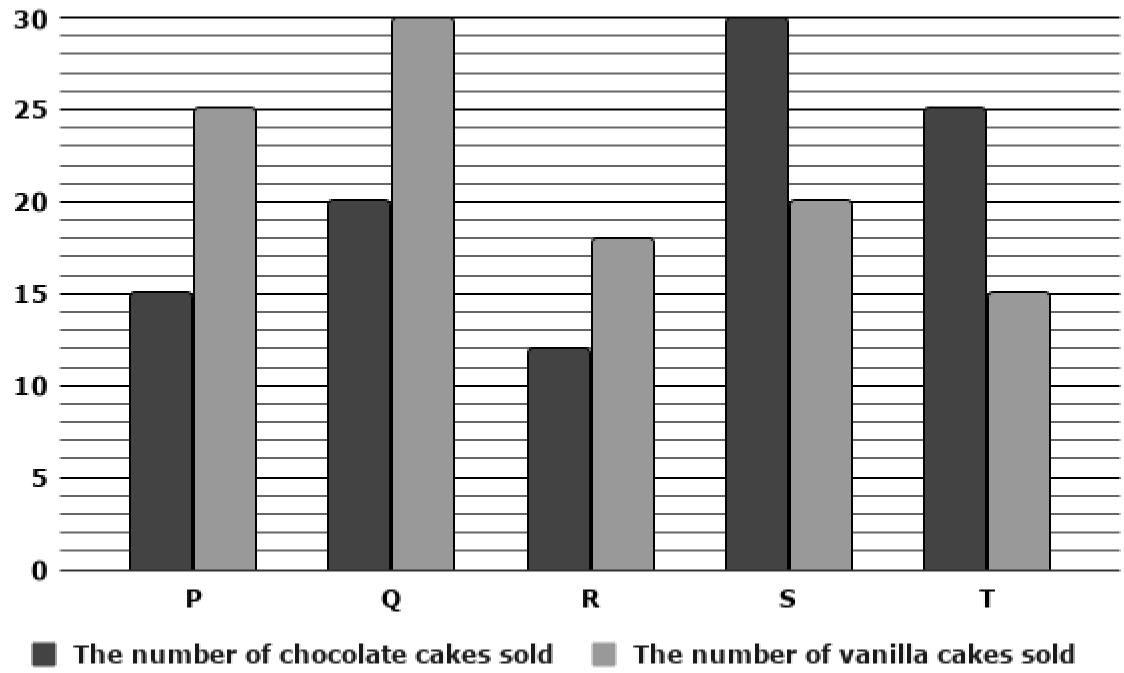
e. 600

6. Questions

Study the following information carefully and answer the questions given below.

The given bar graph shows the number of chocolate cakes and vanilla cakes sold by five different bakeries namely P, Q, R, S and T.

Note: The total number of cakes sold = The number of chocolate cakes sold + The number of vanilla cakes sold.



The total number of cakes sold by bakery P is what percentage more or less than that sold by bakery R?

a. 33.33% less

b. 22.22% more

c. 20% less

d. 33.33% more

e. 25% less

7. Questions

Find the ratio of the number of chocolate cakes sold by bakery R to the number of vanilla cakes sold by bakery T.

a. 3 : 2

b. 4 : 5

c. 2 : 1

d. 5 : 3

e. 2 : 7

8. Questions

Find the total number of chocolate cakes sold by bakeries P, S and T together.

a. 20

b. 50

c. 70

d. 30

e. 40

9. Questions

The number of vanilla cakes sold by bakery S is what percentage of the total number of cakes sold by bakery T?

a. 50%

b. 40%

c. 12%

d. 30%

e. 52%

10. Questions

What is the difference between the total number of vanilla cakes sold by bakeries P, Q and R together and the total number of cakes sold by bakery S?

a. 28

b. 15

c. 20

d. 17

e. 23

11. Questions

Khushal travels from his home to his government office at a speed of 45 km/hr and reaches 12 minutes late. If he increases his speed to 60 km/hr, he reaches the office 8 minutes early. Find the total distance between his home and the office.

a. 60 km

b. 80 km

c. 90 km

d. 70 km

e. 50 km

12. Questions

Priya invested Rs. y in Scheme P and Rs. (y + 3000) in Scheme Q. Scheme P offers compound interest at the rate of 20% p.a compounded annually. If the total interest received by Priya after 2

years from Scheme P is Rs. 2200, find the amount invested by Priya in Scheme Q.

a. Rs. 4800

b. Rs. 8000

c. Rs. 6400

d. Rs. 8400

e. Rs. 7800

13. Questions

A tank can be filled by two inlet pipes P and Q in 12 hours and 15 hours respectively. There is also an outlet pipe M at the bottom of the tank which can empty the full tank in 20 hours. If all three pipes are opened together and the outlet pipe M is closed after 4 hours, in how much additional time will the tank be completely filled?

a. 3 hours

b. 2 hours

c. 4 hours

d. 7 hours

e. 5 hours

14. Questions

A solid wooden cone has a base radius of 14 cm and a slant height of 30 cm. The curved surface of the cone needs to be polished at the rate of Rs. 8 per cm^2. Find the total cost of polishing the curved surface of the cone.

a. 10420

b. 12840

c. 13200

d. 10560

e. 11420

15. Questions

A container contains a mixture of milk and chocolate in the ratio 5 : 3 respectively. If 120 liters of this mixture is taken out and replaced with 25 liters of milk and 30 liters of chocolate, the ratio of milk to chocolate becomes 10 : 7 respectively. Find the initial quantity of chocolate in the mixture.

a. 240 litres

b. 180 litres

c. 250 litres

d. 100 litres

e. 120 litres

16. Questions

S and K started a business together. S invested 40% more than K. After 3 months S withdrew 1/7 of her investment while K increased his capital by 40%. At the end of one year they earned a total profit of Rs. 5100. Find K's share in the profit

a. Rs. 2600

b. Rs. 4120

c. Rs. 4200

d. Rs. 3000

e. Rs. 1500

17. Questions

The ratio of the monthly incomes of Amit and Sumit is 5 : 4 respectively, and the ratio of their monthly expenses is 4 : 3 respectively. If Amit saves Rs. 4000 and Sumit saves Rs. 3500 per month, find the sum of their monthly incomes.

a. Rs. 21000

b. Rs. 18000

c. Rs. 12000

d. Rs. 15000

e. Rs. 24000

18. Questions

The cost price of the item is Rs. 4500. It is sold at a profit of Z%. If the total selling price of the item is Rs. 5400, find the value of Z.

a. 18

b. 10

c. 20

d. 12

e. 15

19. Questions

The present ages of a mother and her daughter are in the ratio 7 : 3 respectively. The difference between their present ages is 28 years. After y years, the mother's age will be twice the age of her daughter. Find the value of y.

a. 2
b. 3
c. 10
d. 7
e. 12

20. Questions

The average score of 6 candidates in an aptitude test is 50. The average score of the first 2 candidates is 45, and the average score of the last 2 candidates is 55. If the score of the 3rd candidate is 10 marks more than the score of the 4th candidate, find the score of the 4th candidate .

a. 79
b. 63
c. 58
d. 72
e. 45

21. Questions

What value should come in the place of (?) in the following questions.

$[(15 * 17) \div 3] + (5)^2 = ?$

a. 140
b. 130
c. 120
d. 110
e. 150

22. Questions

16.5 * 14 + 32 * 6 - 143 = ?

a. 280
b. 130
c. 240
d. 192
e. 231

23. Questions

$(1296 \div \sqrt{36})* 12 + 68 = ?$

a. 2540

b. 2430

c. 2660

d. 2950

e. 2610

24. Questions

$(0.6)^2 * 250 \div 1.5 + 44 = ? * 4$

a. 27

b. 32

c. 28

d. 30

e. 26

25. Questions

23% of $1400 + \sqrt{225} \div 3 - 28.5 = ?$

a. 302.5

b. 298.5

c. 277.5

d. 231.5

e. 245.5

26. Questions

What approximate value will come in the place of (?) in the following questions?

$14.95 * 12.02 + 24.98 * 8.03 - 119.95 = ?$

a. 210

b. 230

c. 260

d. 240

e. 280

27. Questions

$(\sqrt{1023.95} \div 3.99) * 14.98 + (7.95)^2 = ? + 141.92$

a. 42
b. 36
c. 29
d. 45
e. 31

28. Questions

$\sqrt{?} * 11.98 + 159.89 = 399.98 - 120.05$

a. 81
b. 144
c. 121
d. 100
e. 64

29. Questions

$(17.98 * 24.01) \div 8.99 + 256.12 = ?\%$ of 799.91

a. 28
b. 38
c. 39
d. 29
e. 37

30. Questions

$111.02 * 6.98 + (12.99)^2 - 24.95\%$ of $199.88 = ?$

a. 843
b. 729
c. 821
d. 784
e. 896

31. Questions

What value will come in the place of (?) in the following number series?

6, 12, 36, ?, 1260, 13860

a. 200

b. 140

c. 120

d. 180

e. 160

32. Questions

8, 9.7, 13.1, 18.2, ?, 33.5

a. 25

b. 29

c. 22

d. 23

e. 27

33. Questions

4, ?, 12, 39, 160, 805

a. 6

b. 5

c. 7

d. 2

e. 8

34. Questions

324, 307, 288, 267, 244, ?

a. 214

b. 221

c. 215

d. 210

e. 219

35. Questions

?, 27, 36, 52, 77, 113

a. 21
b. 19
c. 23
d. 17
e. 25

36. Questions

Find the wrong number in the following number series.

250, 245, 235, 215, 175, 90, -65

a. 245
b. -65
c. 250
d. 90
e. 175

37. Questions

3, 5, 13, 41, 177, 891, 5353

a. 41
b. 3
c. 5353
d. 13
e. 177

38. Questions

196, 180, 166, 154, 144, 130

a. 130
b. 196
c. 166
d. 154
e. 180

39. Questions

7, 8, 10.5, 21, 52.5, 157.5

a. 21

b. 10.5

c. 8

d. 52.5

e. 7

40. Questions

439, 430, 410, 356, 275, 154

a. 439

b. 410

c. 154

d. 430

e. 275

41. Questions

The following questions contain two equations as I and II. You have to solve both equations and determine the relationship between them and give answer as,

I). $x^2 - x - 20 = 0$

II). $y^2 - 5y - 14 = 0$

a. $x > y$

b. $x \geq y$

c. $x = y$ or relationship cannot be established.

d. $x < y$

e. $x \leq y$

42. Questions

I). $x^2 - 15x + 56 = 0$

II). $y^2 - 13y + 42 = 0$

a. $x > y$

b. $x \geq y$

c. $x = y$ or relationship cannot be established.

d. $x < y$

e. $x \le y$

43. Questions

I). $x^2 + 10x + 24 = 0$

II). $y^2 + 4y + 4 = 0$

a. $x > y$

b. $x \ge y$

c. $x = y$ or relationship cannot be established.

d. $x < y$

e. $x \le y$

44. Questions

I). $x^2 - 14x + 48 = 0$

II). $y^2 - 5y + 6 = 0$

a. $x > y$

b. $x \ge y$

c. $x = y$ or relationship cannot be established.

d. $x < y$

e. $x \le y$

45. Questions

I). $x^2 - 10x + 21 = 0$

II). $y^2 - 17y + 72 = 0$

a. $x > y$

b. $x \ge y$

c. $x = y$ or relationship cannot be established.

d. $x < y$

e. $x \le y$

Explanations:

1. Questions

Store B:

The number of fiction books sold by store B = 250

The number of non-fiction books sold by store B = 250 * 7/5 = 350

The total number of books sold by store B = 250 + 350 = 600

Store D:

The total number of books sold by store D = 600 * 125/100 = 750

The number of non-fiction books sold by store D = 300

The number of fiction books sold by store D = 750 - 300 = 450

Store A:

The number of non-fiction books sold by store A = 250 * 80/100 = 200

The number of fiction books sold by store A = 500 - 200 = 300

The total number of books sold by store A = 500

Store C:

The total number of books sold by store C = 450

The number of non-fiction books sold by store C = 250

The number of fiction books sold by store C = 450 - 250 = 200

Store	The number of fiction books sold	The number of non-fiction books sold	The total number of books sold
A	300	200	500
B	250	350	600
C	200	250	450
D	450	300	750
Total	**1200**	**1100**	**2300**

Answer: D

The number of fiction books sold by store A = 300

The number of non-fiction books sold by store D = 300

The required ratio = 300 : 300 = 1 : 1

Hence, the correct answer is option D.

2. Questions

Store B:

The number of fiction books sold by store B = 250

The number of non-fiction books sold by store B = 250 * 7/5 = 350

The total number of books sold by store B = 250 + 350 = 600

Store D:

The total number of books sold by store D = 600 * 125/100 = 750

The number of non-fiction books sold by store D = 300

The number of fiction books sold by store D = 750 - 300 = 450

Store A:

The number of non-fiction books sold by store A = 250 * 80/100 = 200

The number of fiction books sold by store A = 500 - 200 = 300

The total number of books sold by store A = 500

Store C:

The total number of books sold by store C = 450

The number of non-fiction books sold by store C = 250

The number of fiction books sold by store C = 450 - 250 = 200

Store	The number of fiction books sold	The number of non-fiction books sold	The total number of books sold
A	300	200	500
B	250	350	600
C	200	250	450
D	450	300	750
Total	**1200**	**1100**	**2300**

Answer: B

The total number of books sold by store C = 450

The number of non-fiction books sold by store A = 200

The required percentage = 450 * 100/200 = 225%

Hence, the correct answer is option B.

3. Questions

Store B:

The number of fiction books sold by store B = 250

The number of non-fiction books sold by store B = 250 * 7/5 = 350

The total number of books sold by store B = 250 + 350 = 600

Store D:

The total number of books sold by store D = 600 * 125/100 = 750

The number of non-fiction books sold by store D = 300

The number of fiction books sold by store D = 750 - 300 = 450

Store A:

The number of non-fiction books sold by store A = 250 * 80/100 = 200

The number of fiction books sold by store A = 500 - 200 = 300

The total number of books sold by store A = 500

Store C:

The total number of books sold by store C = 450

The number of non-fiction books sold by store C = 250

The number of fiction books sold by store C = 450 - 250 = 200

Store	The number of fiction books sold	The number of non-fiction books sold	The total number of books sold
A	300	200	500
B	250	350	600
C	200	250	450
D	450	300	750
Total	**1200**	**1100**	**2300**

Answer: E

The average number of books sold by stores A and B together = (500 + 600)/2 = 550

The average number of non-fiction books sold by stores B and C together = (350 + 250)/2 = 300

The required answer = 550 - 300 = 250 more

Hence, the correct answer is option E.

4. Questions

Store B:

The number of fiction books sold by store B = 250

The number of non-fiction books sold by store B = 250 * 7/5 = 350

The total number of books sold by store B = 250 + 350 = 600

Store D:

The total number of books sold by store D = 600 * 125/100 = 750

The number of non-fiction books sold by store D = 300

The number of fiction books sold by store D = 750 - 300 = 450

Store A:

The number of non-fiction books sold by store A = 250 * 80/100 = 200

The number of fiction books sold by store A = 500 - 200 = 300

The total number of books sold by store A = 500

Store C:

The total number of books sold by store C = 450

The number of non-fiction books sold by store C = 250

The number of fiction books sold by store C = 450 - 250 = 200

Store	The number of fiction books sold	The number of non-fiction books sold	The total number of books sold
A	300	200	500
B	250	350	600
C	200	250	450
D	450	300	750
Total	**1200**	**1100**	**2300**

Answer: C

The number of fiction books sold by store B = 250

The number of fiction books sold by store C = 200

x% = (250 - 200) * 100/200

= 25%

x = 25

Hence, the correct answer is option C.

5. Questions

Store B:

The number of fiction books sold by store B = 250

The number of non-fiction books sold by store B = 250 * 7/5 = 350

The total number of books sold by store B = 250 + 350 = 600

Store D:

The total number of books sold by store D = 600 * 125/100 = 750

The number of non-fiction books sold by store D = 300

The number of fiction books sold by store D = 750 - 300 = 450

Store A:

The number of non-fiction books sold by store A = 250 * 80/100 = 200

The number of fiction books sold by store A = 500 - 200 = 300

The total number of books sold by store A = 500

Store C:

The total number of books sold by store C = 450

The number of non-fiction books sold by store C = 250

The number of fiction books sold by store C = 450 - 250 = 200

Store	The number of fiction books sold	The number of non-fiction books sold	The total number of books sold
A	300	200	500
B	250	350	600
C	200	250	450
D	450	300	750
Total	**1200**	**1100**	**2300**

Answer: A

The total number of books sold by store E = 750 * 120/100 = 900

The number of non-fiction books sold by store E = 350 + 50 = 400

The number of fiction books sold by store E = 900 - 400 = 500

Hence, the correct answer is option A.

6. Questions

Bakery P:

The number of chocolate cakes sold by bakery P = 15

The number of vanilla cakes sold by bakery P = 25

The total number of cakes sold by bakery P = 15 + 25 = 40

Bakery Q:

The number of chocolate cakes sold by bakery Q = 20

The number of vanilla cakes sold by bakery Q = 30

The total number of cakes sold by bakery Q = 20 + 30 = 50

Bakery R:

The number of chocolate cakes sold by bakery R = 12

The number of vanilla cakes sold by bakery R = 18

The total number of cakes sold by bakery R = 12 + 18 = 30

Bakery S:

The number of chocolate cakes sold by bakery S = 30

The number of vanilla cakes sold by bakery S = 20

The total number of cakes sold by bakery S = 30 + 20 = 50

Bakery T:

The number of chocolate cakes sold by bakery T = 25

The number of vanilla cakes sold by bakery T = 15

The total number of cakes sold by bakery T = 25 + 15 = 40

Bakery	The number of chocolate cakes sold	The number of vanilla cakes sold	The total number of cakes sold together
P	15	25	40
Q	20	30	50
R	12	18	30
S	30	20	50
T	25	15	40
Total	**102**	**108**	**210**

Answer: D

The total number of cakes sold by bakery P = 40

The total number of cakes sold by bakery R = 30

The required percentage = (40 - 30) * 100/30 = 33.33% more

Hence, the correct answer is option D.

7. Questions

Bakery P:

The number of chocolate cakes sold by bakery P = 15

The number of vanilla cakes sold by bakery P = 25

The total number of cakes sold by bakery P = 15 + 25 = 40

Bakery Q:

The number of chocolate cakes sold by bakery Q = 20

The number of vanilla cakes sold by bakery Q = 30

The total number of cakes sold by bakery Q = 20 + 30 = 50

Bakery R:

The number of chocolate cakes sold by bakery R = 12

The number of vanilla cakes sold by bakery R = 18

The total number of cakes sold by bakery R = 12 + 18 = 30

Bakery S:

The number of chocolate cakes sold by bakery S = 30

The number of vanilla cakes sold by bakery S = 20

The total number of cakes sold by bakery S = 30 + 20 = 50

Bakery T:

The number of chocolate cakes sold by bakery T = 25

The number of vanilla cakes sold by bakery T = 15

The total number of cakes sold by bakery T = 25 + 15 = 40

Bakery	The number of chocolate cakes sold	The number of vanilla cakes sold	The total number of cakes sold together
P	15	25	40
Q	20	30	50
R	12	18	30
S	30	20	50
T	25	15	40
Total	**102**	**108**	**210**

Answer: B

The number of chocolate cakes sold by bakery R = 12

The number of vanilla cakes sold by bakery T = 15

The required ratio = 12 : 15 = 4 : 5

Hence, the correct answer is option B.

8. Questions

Bakery P:

The number of chocolate cakes sold by bakery P = 15

The number of vanilla cakes sold by bakery P = 25

The total number of cakes sold by bakery P = 15 + 25 = 40

Bakery Q:

The number of chocolate cakes sold by bakery Q = 20

The number of vanilla cakes sold by bakery Q = 30

The total number of cakes sold by bakery Q = 20 + 30 = 50

Bakery R:

The number of chocolate cakes sold by bakery R = 12

The number of vanilla cakes sold by bakery R = 18

The total number of cakes sold by bakery R = 12 + 18 = 30

Bakery S:

The number of chocolate cakes sold by bakery S = 30

The number of vanilla cakes sold by bakery S = 20

The total number of cakes sold by bakery S = 30 + 20 = 50

Bakery T:

The number of chocolate cakes sold by bakery T = 25

The number of vanilla cakes sold by bakery T = 15

The total number of cakes sold by bakery T = 25 + 15 = 40

Bakery	The number of chocolate cakes sold	The number of vanilla cakes sold	The total number of cakes sold together
P	15	25	40
Q	20	30	50
R	12	18	30
S	30	20	50
T	25	15	40
Total	**102**	**108**	**210**

Answer: C

The total number of chocolate cakes sold by bakeries P, S and T together = 15 + 30 + 25 = 70

Hence, the correct answer is option C.

9. Questions

Bakery P:

The number of chocolate cakes sold by bakery P = 15

The number of vanilla cakes sold by bakery P = 25

The total number of cakes sold by bakery P = 15 + 25 = 40

Bakery Q:

The number of chocolate cakes sold by bakery Q = 20

The number of vanilla cakes sold by bakery Q = 30

The total number of cakes sold by bakery Q = 20 + 30 = 50

Bakery R:

The number of chocolate cakes sold by bakery R = 12

The number of vanilla cakes sold by bakery R = 18

The total number of cakes sold by bakery R = 12 + 18 = 30

Bakery S:

The number of chocolate cakes sold by bakery S = 30

The number of vanilla cakes sold by bakery S = 20

The total number of cakes sold by bakery S = 30 + 20 = 50

Bakery T:

The number of chocolate cakes sold by bakery T = 25

The number of vanilla cakes sold by bakery T = 15

The total number of cakes sold by bakery T = 25 + 15 = 40

Bakery	The number of chocolate cakes sold	The number of vanilla cakes sold	The total number of cakes sold together
P	15	25	40
Q	20	30	50
R	12	18	30
S	30	20	50
T	25	15	40
Total	**102**	**108**	**210**

Answer: A

The number of vanilla cakes sold by bakery S = 20

The total number of cakes sold by bakery T = 40

The required percentage = 20 * 100/40 = 50%

Hence, the correct answer is option A.

10. Questions

Bakery P:

The number of chocolate cakes sold by bakery P = 15

The number of vanilla cakes sold by bakery P = 25

The total number of cakes sold by bakery P = 15 + 25 = 40

Bakery Q:

The number of chocolate cakes sold by bakery Q = 20

The number of vanilla cakes sold by bakery Q = 30

The total number of cakes sold by bakery Q = 20 + 30 = 50

Bakery R:

The number of chocolate cakes sold by bakery R = 12

The number of vanilla cakes sold by bakery R = 18

The total number of cakes sold by bakery R = 12 + 18 = 30

Bakery S:

The number of chocolate cakes sold by bakery S = 30

The number of vanilla cakes sold by bakery S = 20

The total number of cakes sold by bakery S = 30 + 20 = 50

Bakery T:

The number of chocolate cakes sold by bakery T = 25

The number of vanilla cakes sold by bakery T = 15

The total number of cakes sold by bakery T = 25 + 15 = 40

Bakery	The number of chocolate cakes sold	The number of vanilla cakes sold	The total number of cakes sold together
P	15	25	40
Q	20	30	50
R	12	18	30
S	30	20	50
T	25	15	40
Total	**102**	**108**	**210**

Answer: E

The total number of vanilla cakes sold by bakeries P, Q and R together = 25 + 30 + 18 = 73

The total number of cakes sold by bakery S = 50

The required difference = 73 - 50 = 23

Hence, the correct answer is option E.

11. Questions

Answer: A

Formula used:

When the same distance is traveled at two different speeds:

The total distance = (S_1 * S_2 * time difference)/(speed difference)

The total distance = [(45 * 60) * (20/60)]/(60 - 45)

= [45 * 20] / 15

The total distance = 60 km

Hence, the correct answer is option A.

12. Questions

Answer: B

Formula used:

The compound interest = $P[(1 + R/100)^N - 1]$

The effective rate of CI for 2 years = $(2R + R^2/100)\%$

The effective rate of compound interest = 20 + 20 + 20 * 20/100 = 44%

According to the question,

y * 44% = 2200

y = 5000

The sum invested in scheme Q = 5000 + 3000 = Rs.8000

Hence, the correct answer is option B.

13. Questions

Answer: C

Let the total capacity of the tank = 60 units

The efficiency of pipe P = 60/12 = 5 units per hour

The efficiency of pipe Q = 60/15 = 4 units per hour

The efficiency of pipe M to empty the tank = 60/20 = 3 units per hour

The total capacity of all the pipes = 5 + 4 - 3 = 6 units per hour

The capacity of the tank filled in 4 hours = 6 * 4 = 24 units

The remaining tank = 60 - 24 = 36 units

The time taken by pipe P and pipe Q to fill the tank = 36/9 = 4 hours

Hence, the correct answer is option C.

14. Questions

Answer: D

Formula used:

The curved surface area of cone = πrl

The curved surface area of cone = (22/7) * 14 * 30 = 1320

The total cost of polishing = 1320 * 8 = 10560

Hence, the correct answer is option D.

15. Questions

Answer: E

Let the initial quantity of milk = 5a

The initial quantity of chocolate = 3a

The quantity of milk removed = 120 * 5/8 = 75 litres

The quantity of chocolate removed = 120 * 3/8 = 45 litres

According to the question,

(5a - 75 + 25) / (3a - 45 + 30) = 10 / 7

(5a - 50) / (3a - 15) = 10 / 7

35a - 350 = 30a - 150

5a = 200

a = 40

The initial quantity of chocolate = 3a = 3 * 40 = 120 litres

Hence, the correct answer is option E.

16. Questions

Answer: A

Let the initial investment of K = 5a

The initial investment of S = 5a * 7/5 = 7a

The total investment of K = (5a * 3) + (7a * 9) = 78a

The total investment of S = (7a * 3) + (6a * 9) = 75a

The profit sharing ratio of K and S = 78a : 75a = 26 : 25

K's share of profit = (26 / 51) × 5100 = Rs. 2600.

Hence, the correct answer is option A.

17. Questions

Answer: B

Let the monthly income of Amit = 5a

The income of Sumit = 4a

Let the monthly expenses of Amit = 4b

The monthly expenses of Sumit = 3b

According to the question,

5a - 4b = 4000 -----(1)

4a - 3b = 3500 -----(2)

After solving both equations (1) and (2), we get a = 2000

The total monthly income of Amit and Sumit = 5a + 4a = 9a = 9 * 2000 = Rs. 18000

Hence, the correct answer is option B.

18. Questions

Answer: C

4500 * (100 + Z)/100 = 5400

(100 + Z) = 5400/45

100 + Z = 120

Z = 20

Hence, the correct answer is option C.

19. Questions

Answer: D

Let the present age of mother = 7a

The present age of daughter = 3a

According to the question,

7a - 3a = 28

4a = 28

a = 7

The present age of mother = 7 * 7 = 49 years

The present age of daughter = 7 * 3 = 21 years

And,

(49 + y) / (21 + y) = 2 / 1

49 + y = 42 + 2y

7 = y

Hence, the correct answer is option D.

20. Questions

Answer: E

Total score of all 6 candidates = 6 * 50 = 300.

Total score of the first 2 candidates = 2 * 45 = 90.
Total score of the last 2 candidates = 2 * 55 = 110.

Sum of scores of the first 2 and last 2 candidates = 90 + 110 = 200.

Sum of scores of the 3rd and 4th candidates = 300 - 200 = 100.

Let the score of the 4th candidate be x.

The score of the 3rd candidate = x + 10.

x + (x + 10) = 100
2x + 10 = 100
2x = 90
x = 45.

Hence, the correct answer is option E.

21. Questions

Answer: D

$[(15 * 17) \div 3] + (5)^2 = ?$

$[255 \div 3] + 25 = ?$

85 + 25 = ?

110 = ?

Hence, the correct answer is option D.

22. Questions

Answer: A

16.5 * 14 + 32 * 6 - 143 = ?

231 + 192 - 143 = ?

423 - 143 = ?

280 = ?

Hence, the correct answer is option A.

23. Questions

Answer: C

$(1296 \div \sqrt{36}) * 12 + 68 = ?$

$(1296 \div 6) * 12 + 68 = ?$

2592 + 68 = ?

2660 = ?

Hence, the correct answer is option C.

24. Questions

Answer: E

$(0.6)^2 * 250 \div 1.5 + 44 = ? * 4$

0.36 * 250 ÷ 1.5 + 44 = ? * 4

90 ÷ 1.5 + 44 = ? * 4

60 + 44 = ? * 4

104 = ? * 4

? = 26

Hence, the correct answer is option E.

25. Questions

Answer: B

23% of 1400 + √225 ÷ 3 - 28.5 = ?

322 + 15 ÷ 3 - 28.5 = ?

322 + 5 - 28.5 = ?

327 - 28.5 = ?

298.5 = ?

Hence, the correct answer is option B.

26. Questions

Answer: C

14.95 * 12.02 + 24.98 * 8.03 - 119.95 = ?

15 * 12 + 25 * 8 - 120 = ?

180 + 200 - 120 = ?

260 = ?

Hence, the correct answer is option C.

27. Questions

Answer: A

(√1023.95 ÷ 3.99) * 14.98 + $(7.95)^2$ = ? + 141.92

(√1024 ÷ 4 * 15) + $(8)^2$ = ? + 142

(32 ÷ 4) * 15 + 64 = ? + 142

120 + 64 = ? + 142

184 = ? + 142

42 = ?

Hence, the correct answer is option A.

28. Questions

Answer: D

√? * 11.98 + 159.89 = 399.98 - 120.05

√? * 12 + 160 = 400 - 120

√? * 12 + 160 = 280

√? * 12 = 120

√? = 10

? = 100

Hence, the correct answer is option D.

29. Questions

Answer: B

(17.98 * 24.01) ÷ 8.99 + 256.12 = ?% of 799.91

(18 * 24) ÷ 9 + 256 = ?% of 800

48 + 256 = ? * 8

304 = ? * 8

? = 38

Hence, the correct answer is option B.

30. Questions

Answer: E

111.02 * 6.98 + $(12.99)^2$ - 24.95% of 199.88 = ?

111 * 7 + $(13)^2$ - 25% of 200 = ?

777 + 169 - 50 = ?

946 - 50 = ?

896 = ?

Hence, the correct answer is option E.

31. Questions

Answer: D

6 * 2 = 12

12 * 3 = 36

36 * 5 = 180

180 * 7 = 1260

1260 * 11 = 13860

Missing number: 180

Hence, the correct answer is option D.

32. Questions

Answer: A

8 + 1.7 = 9.7

9.7 + 3.4 = 13.1

13.1 + 5.1 = 18.2

18.2 + 6.8 = 25

25 + 8.5 = 33.5

Missing number: 25

Hence, the correct answer is option A.

33. Questions

Answer: B

4 * 1 + 1 = 5

5 * 2 + 2 = 12

12 * 3 + 3 = 39

39 * 4 + 4 = 160

160 * 5 + 5 = 805

Missing number: 5

Hence, the correct answer is option B.

34. Questions

Answer: E

324 307 288 267 244 219

-17 -19 -21 -23 -25

-2 -2 -2 -2

Missing number: 219

Hence, the correct answer is option E.

35. Questions

Answer: C

$23 + 2^2 = 27$

$27 + 3^2 = 36$

$36 + 4^2 = 52$

$52 + 5^2 = 77$

$77 + 6^2 = 113$

Missing number: 23

Hence, the correct answer is option C.

36. Questions

Answer: D

250 - 5 = 245

245 - 10 = 235

235 - 20 = 215

215 - 40 = 175

175 - 80 = **95**

95 - 160 = -65

Wrong number: 90

Right number: 95

Hence, the correct answer is option D.

37. Questions

Answer: A

3 * 1 + 2 = 5

5 * 2 + 3 = 13

13 * 3 + 4 = **43**

43 * 4 + 5 = 177

177 * 5 + 6 = 891

891 * 6 + 7 = 5353

Wrong number: 41

Right number: 43

Hence, the correct answer is option A.

38. Questions

Answer: A

196 180 166 154 144 **136**

-16 -14 -12 -10 -8

+2 +2 +2 +2

Wrong number: 130

Right number: 136

Hence, the correct answer is option A.

39. Questions

Answer: C

7 * 1 = **7**

7 * 1.5 = 10.5

10.5 * 2 = 21

21 * 2.5 = 52.5

52.5 * 3 = 157.5

Wrong number: 8

Right number: 7

Hence, the correct answer is option C.

40. Questions

Answer: B

439 - 3^2 = 430

430 - 5^2 = **405**

405 - 7^2 = 356

356 - 9^2 = 275

275 - 11^2 = 154

Wrong number: 410

Right number: 405

Hence, the correct answer is option B.

41. Questions

Answer: C

I). x^2 - x - 20 = 0

x^2 - 5x + 4x - 20 = 0

x (x - 5) + 4 (x - 5) = 0

(x + 4) (x - 5) = 0

x = 5 or -4

II). y^2 - 5y - 14 = 0

y^2 - 7y + 2y - 14 = 0

y (y - 7) + 2 (y - 7) = 0

(y + 2) (y - 7) = 0

y = 7 or -2

So, x = y or relationship cannot be established.

Hence, the correct answer is option C.

42. Questions

Answer: B

I). x^2 - 15x + 56 = 0

x^2 - 8x - 7x + 56 = 0

x (x - 8) - 7 (x - 8) = 0

(x - 7) (x - 8) = 0

x = 7 or 8

II). y^2 - 13y + 42 = 0

y^2 - 7y - 6y + 42 = 0

y (y - 7) - 6 (y - 7) = 0

(y - 7) (y - 6) = 0

y = 7 or 6

So, x ≥ y

Hence, the correct answer is option B.

43. Questions

Answer: D

I). x^2 + 10x + 24 = 0

x^2 + 4x + 6x + 24 = 0

x (x + 4) + 6 (x + 4) = 0

$(x + 6)(x + 4) = 0$

$x = -6$ or -4

II). $y^2 + 4y + 4 = 0$

$y^2 + 2y + 2y + 4 = 0$

$y(y + 2) + 2(y + 2) = 0$

$(y + 2)(y + 2) = 0$

$y = -2$ or -2

So, $x < y$

Hence, the correct answer is option D.

44. Questions

Answer: A

I). $x^2 - 14x + 48 = 0$

$x^2 - 8x - 6x + 48 = 0$

$x(x - 8) - 6(x - 8) = 0$

$(x - 6)(x - 8) = 0$

$x = 6$ or 8

II). $y^2 - 5y + 6 = 0$

$y^2 - 3y - 2y + 6 = 0$

$y(y - 3) - 2(y - 3) = 0$

$(y - 2)(y - 3) = 0$

$y = 2$ or 3

So, $x > y$

Hence, the correct answer is option A.

45. Questions

Answer: D

I). $x^2 - 10x + 21 = 0$

$x^2 - 7x - 3x + 21 = 0$

$x(x - 7) - 3(x - 7) = 0$

$(x - 3)(x - 7) = 0$

x = 3 or 7

II). $y^2 - 17y + 72 = 0$

$y^2 - 9y - 8y + 72 = 0$

$y (y - 9) - 8 (y - 9) = 0$

$(y - 8) (y - 9) = 0$

y = 8 or 9

So, $x < y$

Hence, the correct answer is option D.